

October 3, 2026

The Editor-in-Chief

Applied Intelligence (Springer Nature)

Subject: Submission of an Original Research article — “Sparsity-Adaptive Gated Eviction for Memory-Efficient Long-Context Inference of Large Language Models”

Dear Editor,

I submit the manuscript “Sparsity-Adaptive Gated Eviction for Memory-Efficient Long-Context Inference of Large Language Models” for consideration as an Original Research article in *Applied Intelligence*.

Research problem. Long-context inference of large language models is bounded by the key-value (KV) cache, which grows linearly with context and exhausts accelerator memory well before compute does. Existing training-free eviction methods lose quality quickly once the cache is reduced below roughly half of its full size, which prevents aggressive memory savings in deployment.

Method. The manuscript proposes SAGE, a training-free framework that formulates KV cache eviction as a global memory-contract constrained allocation problem and couples three mechanisms under that single contract: an exponentially decayed attention-confidence score that tracks the current relevance of each cached entry, a layer-adaptive budget allocator that re-allocates cache capacity online according to measured attention saturation, and a dual-reservoir structure that protects the recent window and attention sinks while evicting only from the history region.

Main contributions. (i) The unified memory-contract formulation with online, entropy-driven layer budgets, which no cited prior component provides; (ii) the decayed-confidence scoring rule with the protected reservoir, whose separate and combined values are priced in a one-at-a-time ablation; (iii) an evaluation covering language quality, long-context tasks, retrieval to 128K tokens, and wall-clock efficiency on LLaMA-2-7B/13B-Chat and Vicuna-13B.

Key findings. At a nominal 20% cache budget, SAGE holds WikiText-2 perplexity within 0.34 points of the full cache (5.46 against 5.12 on the 7B model), retains 98.0% of the LongBench average, and reaches 94.8% needle-retrieval accuracy at a 128K context. The realized footprint is 22.6% of the full cache including all bookkeeping. At 32K tokens the KV memory shrinks by a factor of 4.4 and decode throughput rises by $1.52\times$, growing to $2.02\times$ at 128K, which extends the context capacity of a 24 GB GPU from 17K to 74K tokens.

Relevance to Applied Intelligence. The work targets the applied side of intelligent systems: it is a deployable, training-free mechanism that converts a hard memory constraint into a controllable quality-memory-throughput trade-off, and its contribution is an efficient inference technique for large language models running under practical hardware limits. I believe this fits the journal’s scope of original intelligent-systems methodology addressing resource-constrained real-world deployment.

Originality and exclusivity. The manuscript is original, has not been published previously in any form, and is not under consideration by any other journal. All references were resolved and cross-checked programmatically against the arXiv, Crossref, and OpenAlex records before submission.

Competing interests. The author declares no competing financial or non-financial interests. No funding was received for this work.

Corresponding author.

Name	Chang Tan (谭畅)
Position	Undergraduate Student
Affiliation	School of Economics and Management, Dalian University of Technology
Address	No. 2 Linggong Road, Ganjingzi District, Dalian 116024, Liaoning, China
E-mail	aquars43@foxmail.com

Chang Tan
Sole and corresponding author

Thank you for your consideration.

Sincerely,

Chang Tan